MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3DP1A AND 3DP1A+S2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus with decalcomania inscription on tube type 3DP1A+S2

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	Vdc	ft
Maximum:	6.9	0	550	1,100	2,200	---	1.5	1.0	-125	10,000
Minimum:	5.7	-200	---	---	1,500	7.5	---	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	1,500	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(B) 5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	2.25	Inches
(B) 5216	Cathode illumination	-		----	---	---	---
5248	Deflection-factor uniformity	-		----	---	---	---
1331	Direct-interelectrode capacitance	-	Control grid to all	Cg1	---	10.5	pF
			Cathode to all	Ck	---	10.5	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	4.0	pF
			D1 to all	CD1	---	15.0	pF
			D3 to all	CD3	---	12.0	pF
			D1 to all, except D2	CD1	---	13.0	pF
			D2 to all, except D1	CD2	---	13.0	pF
			D3 to all, except D4	CD3	---	12.0	pF
			D4 to all, except D3	CD4	---	12.0	pF
			D5 to anode 2	CD5	---	3.5	pF
(B) 1216	Base material insulating quality	-	Zone 5 (min)	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1,2	Light = 7.5 ftL	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
(B) 5221	Light output	-		Light	7.5	---	ftL
5223	Modulation	-	Light = 7.5 ftL	Δfc	---	40	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	15	mm
5231	Spot displacement (leakage)	-		Displ	---	5	mm
5241	Grid cutoff voltage	-		Ec1	-22.5	-67.5	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-50	10	μAdc
(B) 5201	Electrode current (cathode)	-	Light = 7.5 ftL	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	-	+1D2, pin No. 5	---	---	---	---
5101	Angle between traces	-		---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 2,200 Vdc	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5226	Line width "A" (electrostatic deflection)	3,5	Light = 7.5 ftL	Width	---	0.60	mm
5226	Line width "B" (electrostatic deflection)	3,6	Light = 7.5 ftL	Width	---	0.80	mm
5246	Focusing voltage at cutoff	-		Eb1	345	515	Vdc
(B) 5246	Focusing voltage (zero bias)	-		Eb1	300	515	Vdc
5248	Deflection factor	-	1D2	DF	127	173	Vdc/in.
5248	Deflection factor	-	3D4	DF	As recorded		Vdc/in.
---	Deflection factor	4	D5	Pk to Pk	50	84	Vdc
---	Radial pattern width	4		Width	---	10	mm
---	Deflection ratio	-	1D2/3D4	Ratio	1.14	1.56	---
5251	Heater-cathode leakage current	-		---	---	---	---
(B) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 1						
---	Life test	-	Group C, light = 7.5 ftL; Eb2 = 2,200 Vdc, t = 500 hours	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Light = 7.5 ftL	ΔEc	---	50	Vdc
5226	Line width "A"	-	Light = 7.5 ftL	Width	---	0.70	mm
5226	Line width "B"	-	Light = 7.5 ftL	Width	---	1.05	mm

NOTES:

1. The center of the "gas cross" appearing inside pattern will not appear due to presence of rod in tube face.
2. This test to be performed at the conclusion of the holding period.
3. Modify the regular procedure of line width measurements in the following manner: Adjustment shall be made for best focus at a point 1/2 inch (12.70 mm) from the center of the radial deflecting electrode terminal along the line of the high-frequency scan.
4. Using mask centered with respect to tube face, set diameter of scanned circle to 1-3/8 inch (34.93 mm) and remove mask. Apply and adjust 35 KHz signal on D5 (rod) so that minimum width of annular pattern is 5 mm. The maximum radial width of same pattern shall be within the limit specified. The peak voltage shall be within the limits specified.
5. The tubes shall be observed for deflection defocusing, astigmatism and spot ellipticity observable to the eye, as evidenced by fuzziness due to lack of sharpness of trace (usually around the edges), bow-tieing (irregular widths of any line when observed at different points) and bowing of trace other than that normally caused by curvatures of the bulb.
6. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 5.

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NOTES: -Continued

7. The following applies to tube type 3DP1A+S2.

(a) Before application of the inscription, the tube shall have passed all requirements of this specification sheet.
(b) The inscription outlined on figure 2 shall be secured to the tube and be positioned in the following manner:
(1) The trace produced by deflecting electrodes 1D2 shall not deviate by more than 2.5 degrees from a line passing through scale markings 0 and 2.5 on the inscription.
(2) The position of zero on the inscription shall be 64.28 degrees ±12.5 degrees clockwise from the centerline of the base lug key, looking at the tube face.
(3) The inscription shall be centered with respect to the tube face within 3/64 (1.19 mm) of an inch
(4) Mask center electrode.
(5) After masking, the tube face shall be sprayed with a light coat of clear lacquer.

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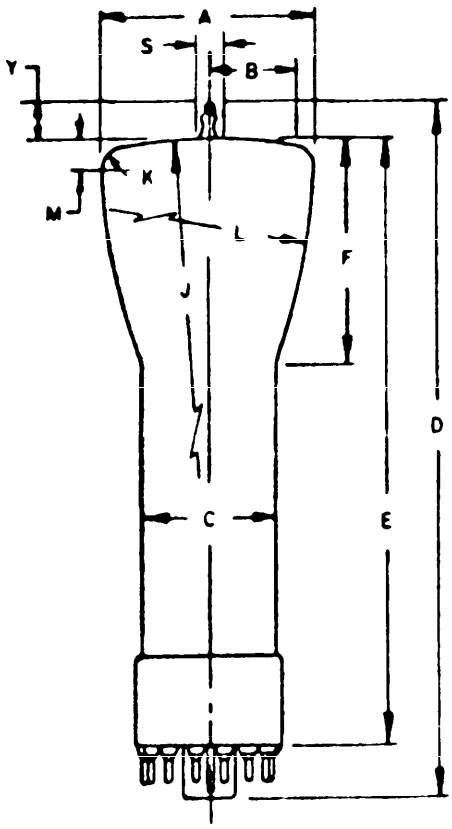
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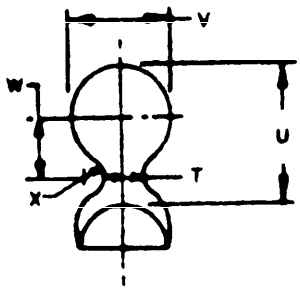
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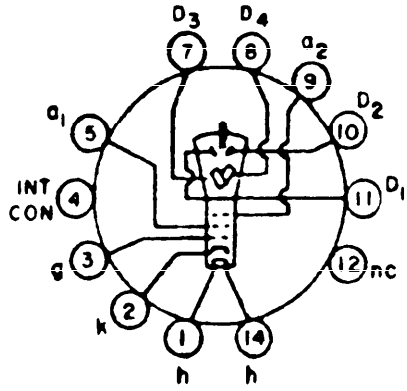
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.937 (74.60)	3.063 (77.80)
B	See note a	
C	1.937 (49.20)	2.063 (52.40)
D	10.063 (255.60)	10.750 (273.08)
S	See note b	
U	.141 (3.58)	---
V	.110 (2.79)	.116 (2.95)
Y	.313 (7.95)	.501 (12.73)
Reference dimensions (see note f)		
E	9.250 (234.95)	
F	3.875 (98.43)	
J	8.000 (203.20) R	
K	.250 (6.35) R	
L	12.438 (315.93) R	
M	.350 (8.89)	
T	.068 (1.73) Dia	
W	.058 (1.47)	
X	.030 (.76) R	





NOTES:
a. The minimum useful screen radius shall not be less than 1.375 (34.93 mm).
b. The maximum diameter uncoated area around center of the radial deflecting electrode terminal shall not be more than .625 (15.88 mm).
c. The radial deflecting electrode terminal shall be eccentric with respect to the tube axis by not more than .063 (1.60 mm).
d. The base shall be a medium shell DIHEPTAL 12-pin (B12-37) type.
e. The bulb shall be a J24G type.
f. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3DP1A AND 3DP1A+S2.

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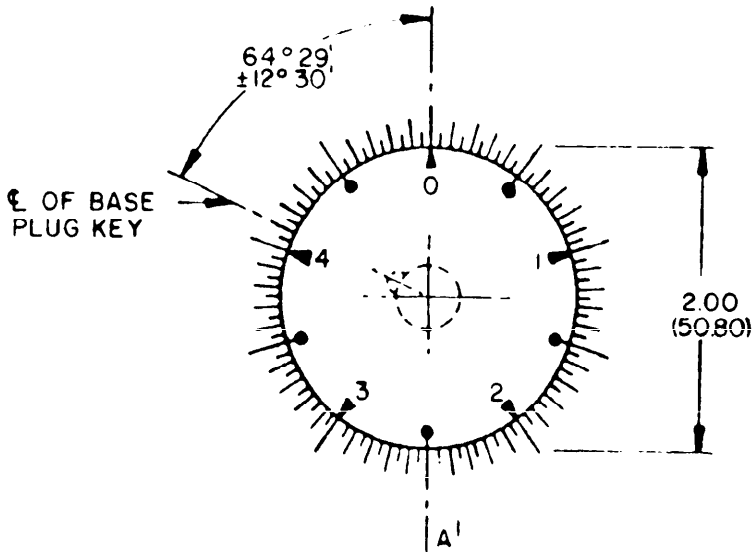


FIGURE 2. S2 scale.

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